

Electronic Supplementary Information

Machine Learning-Guided Prediction of Ionizable Amphiphilic Janus Dendrimers for mRNA Nanomedicine

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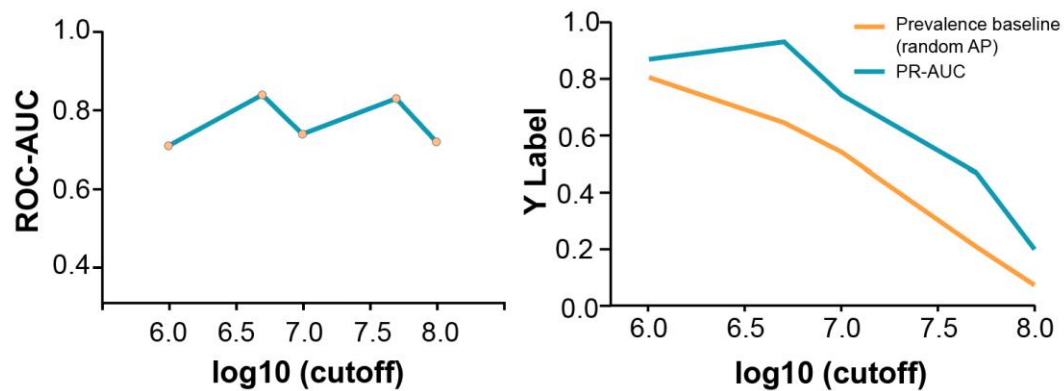


Fig. S1 Sensitivity of classification performance to the binarization threshold used for labeling IAJDs.

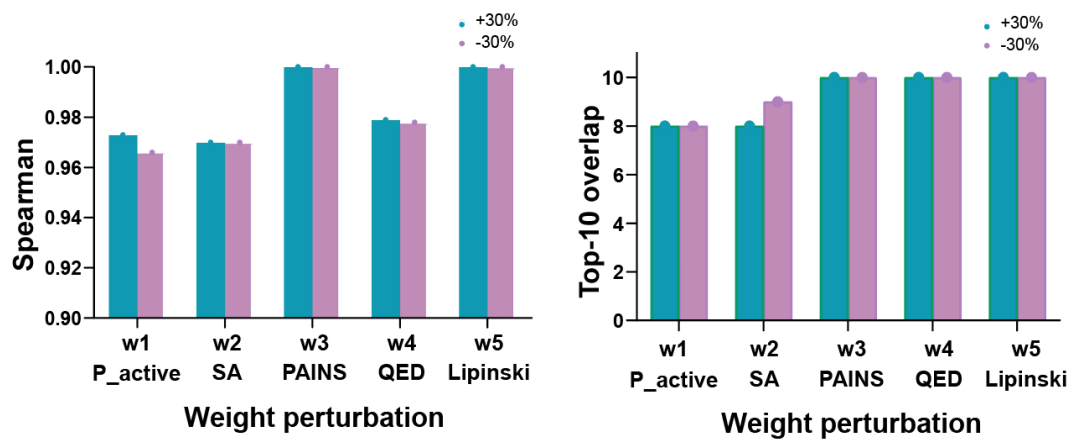


Fig. S2 Robustness of candidate ranking to perturbations in the multi-objective scoring weights.

Table S1 Dataset of IAJD molecules with physicochemical properties and labels.

Name	MolWt	LogP	TPSA	HBA	HBD	RotB	Aromatic Rings	FractionCSP3	label	Results
IAJD_1	1165.598	11.6298	178.63	17	1	61	2	0.78125	0	2.395
IAJD_2	1165.598	11.6298	178.63	17	1	61	2	0.78125	0	0.449
IAJD_3	1278.758	11.9109	198.94	19	1	67	2	0.785714286	0	0
IAJD_4	1292.785	11.9534	198.94	19	1	69	2	0.788732394	0	0
IAJD_5	1363.864	12.107	219.25	21	1	69	2	0.783783784	0	0
IAJD_6	1331.821	14.8131	178.63	17	1	67	4	0.662337662	0	0
IAJD_7	1331.821	14.8131	178.63	17	1	67	4	0.662337662	0	0
IAJD_8	1368.883	13.5238	198.94	19	1	71	3	0.727272727	0	0.3959
IAJD_9a	1368.883	13.5238	198.94	19	1	71	3	0.727272727	1	21.6
IAJD_19	1256.667	10.403	198.94	19	1	63	3	0.695652174	0	8.778
IAJD_20	1284.721	11.1832	198.94	19	1	65	3	0.704225352	0	0.9772
IAJD_21	1312.775	11.9634	198.94	19	1	67	3	0.712328767	0	0
IAJD_22	1340.829	12.7436	198.94	19	1	69	3	0.72	0	0.5814
IAJD_23	1256.667	10.1148	198.94	19	1	61	3	0.695652174	0	5.555
IAJD_24	1312.775	11.387	198.94	19	1	63	3	0.712328767	0	0.4195
IAJD_25	1553.206	17.8235	208.17	20	1	83	3	0.764044944	0	0.6829
IAJD_30	983.555	14.1048	113.07	11	0	54	0	0.948275862	0	1.887
IAJD_31	1464.065	16.5325	205.37	21	0	79	2	0.817073171	1	47.67
IAJD_33a	1420.959	15.2874	198.94	19	1	67	3	0.740740741	1	144.3
IAJD_34	1392.905	14.5072	198.94	19	1	65	3	0.734177215	1	150.7
IAJD_35	1450.989	12.8104	205.42	21	1	67	3	0.740740741	0	0
IAJD_36	1422.935	12.0302	205.42	21	1	65	3	0.734177215	0	0
IAJD_37	1544.195	18.3811	205.37	21	0	79	2	0.829545455	0	7.767
IAJD_44	1205.663	12.5541	178.63	17	1	61	2	0.791044776	1	47.67
IAJD_45	1220.678	11.3156	181.87	18	1	61	2	0.791044776	0	0.8887
IAJD_64	589.946	10.671	48	5	0	30	1	0.810810811	0	0
IAJD_65	630.011	11.5953	48	5	0	30	1	0.825	0	0
IAJD_66	645.026	10.3568	51.24	6	0	30	1	0.825	1	25.8
IAJD_70	644.042	9.9298	54.04	5	1	30	1	0.825	0	6.62
IAJD_71	675.052	9.7193	71.47	7	1	32	1	0.829268293	1	37.9
IAJD_73	616.972	9.5766	51.24	6	0	28	1	0.815789474	0	NaN
IAJD_74	701.134	11.9172	51.24	6	0	34	1	0.840909091	1	17.1
IAJD_75	757.242	13.4776	51.24	6	0	38	1	0.854166667	1	33.7
IAJD_76	532.81	6.9478	51.24	6	0	20	1	0.78125	1	78.1
IAJD_77	560.864	8.0162	51.24	6	0	24	1	0.794117647	0	2.44
IAJD_78	645.026	10.3568	51.24	6	0	30	1	0.825	0	0.112
IAJD_79	829.349	14.6565	60.47	7	0	42	1	0.865384615	1	13.4
IAJD_81	715.161	12.3073	51.24	6	0	35	1	0.844444444	1	23.2
IAJD_82	809.359	13.3561	60.47	7	0	45	0	0.98	0	1.62
IAJD_83	839.385	12.7186	80.7	8	1	47	0	0.980392157	1	71.1
IAJD_84	630.999	9.9667	51.24	6	0	29	1	0.820512821	0	0.689

IAJD_85	704.99	7.8828	103.84	10	0	30	1	0.775	0	2.87
IAJD_86	616.972	9.5766	51.24	6	0	28	1	0.815789474	1	47.3
IAJD_87	562.836	6.3103	71.47	7	1	22	1	0.787878788	1	32.4
IAJD_88	532.81	7.236	51.24	6	0	22	1	0.78125	0	2.34
IAJD_89	562.836	6.5985	71.47	7	1	24	1	0.787878788	1	59.2
IAJD_90	588.918	8.7964	51.24	6	0	26	1	0.805555556	0	NaN
IAJD_91	618.944	8.1589	71.47	7	1	28	1	0.810810811	0	9.94
IAJD_92	641.035	8.2426	60.47	7	0	30	0	0.973684211	0	NaN
IAJD_93	671.061	7.6051	80.7	8	1	32	0	0.974358974	1	426
IAJD_95	763.07	7.7373	124.07	11	1	32	1	0.790697674	1	66.1
IAJD_96	661.025	9.543	60.47	7	0	27	1	0.825	1	47.9
IAJD_97	691.051	8.9055	80.7	8	1	29	1	0.829268293	1	58.9
IAJD_98	813.35	15.038	51.24	6	0	42	1	0.865384615	0	2.06
IAJD_99	843.376	14.4005	71.47	7	1	44	1	0.867924528	0	4.49
IAJD_100	977.683	18.0373	60.47	7	0	57	0	0.983870968	0	NaN
IAJD_101	1007.709	17.3998	80.7	8	1	59	0	0.984126984	0	NaN
IAJD_103	532.81	6.9478	51.24	6	0	20	1	0.78125	0	1.7
IAJD_105	659.053	10.7469	51.24	6	0	31	1	0.829268293	1	47.4
IAJD_106	701.134	11.9172	51.24	6	0	34	1	0.840909091	1	34.8
IAJD_107	729.188	12.6974	51.24	6	0	36	1	0.847826087	1	21.9
IAJD_108	700.15	11.4902	54.04	5	1	34	1	0.840909091	0	9.66
IAJD_110	1421.943	15.7144	196.14	20	0	67	3	0.740740741	1	88.5
IAJD_111	1393.889	14.9342	196.14	20	0	65	3	0.734177215	1	72.1
IAJD_112	574.891	8.4063	51.24	6	0	25	1	0.8	0	4.03
IAJD_113	604.917	7.7688	71.47	7	1	27	1	0.805555556	1	17
IAJD_116	645.026	10.3568	51.24	6	0	30	1	0.825	1	23.8
IAJD_117	675.052	9.7193	71.47	7	1	32	1	0.829268293	0	8.64
IAJD_119	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	82.2
IAJD_120	717.133	10.8896	71.47	7	1	35	1	0.840909091	1	24.8
IAJD_121	588.918	8.7964	51.24	6	0	26	1	0.805555556	0	NaN
IAJD_122	618.944	8.1589	71.47	7	1	28	1	0.810810811	1	25
IAJD_123	630.999	9.9667	51.24	6	0	29	1	0.820512821	0	7.23
IAJD_124	661.025	9.3292	71.47	7	1	31	1	0.825	1	37.7
IAJD_125	689.079	10.1094	71.47	7	1	33	1	0.833333333	1	215
IAJD_127	731.16	11.2797	71.47	7	1	36	1	0.844444444	1	107
IAJD_128	745.187	11.6698	71.47	7	1	37	1	0.847826087	1	55.3
IAJD_129	574.891	8.2622	51.24	6	0	24	1	0.8	0	1.65
IAJD_130	604.917	7.6247	71.47	7	1	26	1	0.805555556	1	118
IAJD_131	1492.078	17.6649	196.14	20	0	72	3	0.755813953	0	1.86
IAJD_133	658.069	10.3199	54.04	5	1	31	1	0.829268293	1	11.1
IAJD_134	1205.663	12.5541	178.63	17	1	61	2	0.791044776	1	26.7
IAJD_135	630.999	9.8226	51.24	6	0	28	1	0.820512821	1	12.2
IAJD_136	661.025	9.1851	71.47	7	1	30	1	0.825	1	10.4
IAJD_137	630.999	9.9667	51.24	6	0	29	1	0.820512821	0	3.54

IAJD_138	661.025	9.3292	71.47	7	1	31	1	0.825	1	14.9
IAJD_139	645.026	10.3568	51.24	6	0	30	1	0.825	0	1.67
IAJD_140	675.052	9.7193	71.47	7	1	32	1	0.829268293	0	5.9
IAJD_141	659.053	10.7469	51.24	6	0	31	1	0.829268293	1	20.6
IAJD_142	689.079	10.1094	71.47	7	1	33	1	0.833333333	1	25.1
IAJD_143	687.107	11.5271	51.24	6	0	33	1	0.837209302	1	12.8
IAJD_144	717.133	10.8896	71.47	7	1	35	1	0.840909091	1	13.9
IAJD_147	715.161	12.3073	51.24	6	0	35	1	0.844444444	0	9.24
IAJD_148	745.187	11.6698	71.47	7	1	37	1	0.847826087	1	63.3
IAJD_149	743.215	13.0875	51.24	6	0	37	1	0.85106383	1	22.6
IAJD_150	773.241	12.45	71.47	7	1	39	1	0.854166667	1	34.7
IAJD_151	757.242	13.4776	51.24	6	0	38	1	0.854166667	1	31.2
IAJD_152	787.268	12.8401	71.47	7	1	40	1	0.857142857	1	19.7
IAJD_153	771.269	13.8677	51.24	6	0	39	1	0.857142857	1	73.3
IAJD_154	801.295	13.2302	71.47	7	1	41	1	0.86	1	23.9
IAJD_155	1351.808	13.6198	196.14	20	0	61	3	0.723684211	1	101
IAJD_156	1435.97	16.1045	196.14	20	0	68	3	0.743902439	0	5.83
IAJD_157	1449.997	16.4946	196.14	20	0	69	3	0.746987952	1	36.6
IAJD_158	1478.051	17.2748	196.14	20	0	71	3	0.752941176	1	12.3
IAJD_159	1477.067	16.8478	198.94	19	1	71	3	0.752941176	1	68.6
IAJD_161	602.945	9.0424	51.24	6	0	26	1	0.810810811	1	84.8
IAJD_162	632.971	8.4049	71.47	7	1	28	1	0.815789474	1	203
IAJD_163	602.945	9.1865	51.24	6	0	27	1	0.810810811	0	1.9
IAJD_164	632.971	8.549	71.47	7	1	29	1	0.815789474	1	32
IAJD_165	616.972	9.5766	51.24	6	0	28	1	0.815789474	0	NaN
IAJD_166	646.998	8.9391	71.47	7	1	30	1	0.820512821	0	NaN
IAJD_167	630.999	9.9667	51.24	6	0	29	1	0.820512821	0	6.6
IAJD_168	661.025	9.3292	71.47	7	1	31	1	0.825	0	NaN
IAJD_169	659.053	10.7469	51.24	6	0	31	1	0.829268293	0	2.1
IAJD_170	689.079	10.1094	71.47	7	1	33	1	0.833333333	0	NaN
IAJD_171	687.107	11.5271	51.24	6	0	33	1	0.837209302	1	76.8
IAJD_172	717.133	10.8896	71.47	7	1	35	1	0.840909091	1	68.2
IAJD_173	715.161	12.3073	51.24	6	0	35	1	0.844444444	1	123
IAJD_174	745.187	11.6698	71.47	7	1	37	1	0.847826087	0	NaN
IAJD_175	729.188	12.6974	51.24	6	0	36	1	0.847826087	0	NaN
IAJD_176	759.214	12.0599	71.47	7	1	38	1	0.85106383	0	NaN
IAJD_177	743.215	13.0875	51.24	6	0	37	1	0.85106383	1	91.8
IAJD_178	773.241	12.45	71.47	7	1	39	1	0.854166667	1	405
IAJD_179	546.837	7.482	51.24	6	0	22	1	0.787878788	1	11
IAJD_180	576.863	6.8445	71.47	7	1	24	1	0.794117647	1	108
IAJD_181	546.837	7.6261	51.24	6	0	23	1	0.787878788	1	14.3
IAJD_182	576.863	6.9886	71.47	7	1	25	1	0.794117647	1	35
IAJD_183	574.891	8.4063	51.24	6	0	25	1	0.8	1	36.4
IAJD_184	604.917	7.7688	71.47	7	1	27	1	0.805555556	0	7.68

IAJD_185	602.945	9.1865	51.24	6	0	27	1	0.810810811	0	1.42
IAJD_186	632.971	8.549	71.47	7	1	29	1	0.815789474	1	152
IAJD_187	630.999	9.9667	51.24	6	0	29	1	0.820512821	0	6.01
IAJD_188	661.025	9.3292	71.47	7	1	31	1	0.825	1	62.4
IAJD_189	659.053	10.7469	51.24	6	0	31	1	0.829268293	1	12.3
IAJD_190	689.079	10.1094	71.47	7	1	33	1	0.833333333	1	67.4
IAJD_191	673.08	11.137	51.24	6	0	32	1	0.833333333	1	11.7
IAJD_192	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	31.3
IAJD_193	687.107	11.5271	51.24	6	0	33	1	0.837209302	0	4.43
IAJD_194	717.133	10.8896	71.47	7	1	35	1	0.840909091	1	32.8
IAJD_197	588.918	8.6523	51.24	6	0	25	1	0.805555556	0	4.8
IAJD_198	618.944	8.0148	71.47	7	1	27	1	0.810810811	1	65.3
IAJD_199	588.918	8.7964	51.24	6	0	26	1	0.805555556	0	1.96
IAJD_200	618.944	8.1589	71.47	7	1	28	1	0.810810811	1	71.8
IAJD_201	646.998	8.9391	71.47	7	1	30	1	0.820512821	1	59.3
IAJD_202	673.08	11.137	51.24	6	0	32	1	0.833333333	0	3.69
IAJD_203	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	41.5
IAJD_204	731.16	11.2797	71.47	7	1	36	1	0.844444444	0	4.58
IAJD_205	715.161	12.3073	51.24	6	0	35	1	0.844444444	1	10.8
IAJD_206	745.187	11.6698	71.47	7	1	37	1	0.847826087	1	65
IAJD_207	759.214	12.0599	71.47	7	1	38	1	0.85106383	1	18.3
IAJD_208	616.972	9.4325	51.24	6	0	27	1	0.815789474	0	2.85
IAJD_209	646.998	8.795	71.47	7	1	29	1	0.820512821	0	8.5
IAJD_210	616.972	9.5766	51.24	6	0	28	1	0.815789474	0	4.1
IAJD_211	646.998	8.9391	71.47	7	1	30	1	0.820512821	1	127
IAJD_212	645.026	10.3568	51.24	6	0	30	1	0.825	0	7.6
IAJD_213	675.052	9.7193	71.47	7	1	32	1	0.829268293	1	28
IAJD_214	757.242	13.4776	51.24	6	0	38	1	0.854166667	1	11.7
IAJD_215	787.268	12.8401	71.47	7	1	40	1	0.857142857	0	1.5
IAJD_216	645.026	10.2127	51.24	6	0	29	1	0.825	1	15.4
IAJD_217	675.052	9.5752	71.47	7	1	31	1	0.829268293	1	96
IAJD_218	645.026	10.3568	51.24	6	0	30	1	0.825	0	2.08
IAJD_219	675.052	9.7193	71.47	7	1	32	1	0.829268293	0	5.4
IAJD_220	673.08	11.137	51.24	6	0	32	1	0.833333333	0	5.08
IAJD_221	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	115
IAJD_222	729.188	12.6974	51.24	6	0	36	1	0.847826087	0	7.34
IAJD_223	759.214	12.0599	71.47	7	1	38	1	0.85106383	1	11.6
IAJD_224	771.269	13.8677	51.24	6	0	39	1	0.857142857	1	45.8
IAJD_225	801.295	13.2302	71.47	7	1	41	1	0.86	1	54.3
IAJD_226	785.296	14.2578	51.24	6	0	40	1	0.86	1	32
IAJD_227	815.322	13.6203	71.47	7	1	42	1	0.862745098	1	13.1
IAJD_228	659.053	10.6028	51.24	6	0	30	1	0.829268293	1	13
IAJD_229	689.079	9.9653	71.47	7	1	32	1	0.833333333	1	41.7
IAJD_230	659.053	10.7469	51.24	6	0	31	1	0.829268293	1	16.8

IAJD_231	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	48.2
IAJD_232	687.107	11.5271	51.24	6	0	33	1	0.837209302	1	29.8
IAJD_233	717.133	10.8896	71.47	7	1	35	1	0.840909091	1	98.1
IAJD_234	743.215	13.0875	51.24	6	0	37	1	0.85106383	0	2.51
IAJD_235	773.241	12.45	71.47	7	1	39	1	0.854166667	1	73.6
IAJD_236	673.08	10.9929	51.24	6	0	31	1	0.833333333	0	3.77
IAJD_237	703.106	10.3554	71.47	7	1	33	1	0.837209302	1	14.9
IAJD_238	673.08	11.137	51.24	6	0	32	1	0.833333333	1	10.1
IAJD_239	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	30.9
IAJD_240	701.134	11.9172	51.24	6	0	34	1	0.840909091	0	5.97
IAJD_241	731.16	11.2797	71.47	7	1	36	1	0.844444444	1	11.9
IAJD_242	799.323	14.6479	51.24	6	0	41	1	0.862745098	1	1.5
IAJD_243	829.349	14.0104	71.47	7	1	43	1	0.865384615	0	1.4
IAJD_244	560.864	8.0162	51.24	6	0	24	1	0.794117647	0	9.21
IAJD_245	590.89	7.3787	71.47	7	1	26	1	0.8	1	60.9
IAJD_247	560.864	7.8721	51.24	6	0	23	1	0.794117647	1	35.4
IAJD_248	641.035	8.6749	60.47	7	0	33	0	0.973684211	0	2
IAJD_249	671.061	8.0374	80.7	8	1	35	0	0.974358974	1	597
IAJD_250	767.278	12.1858	60.47	7	0	42	0	0.978723404	0	NaN
IAJD_251	797.304	11.5483	80.7	8	1	44	0	0.979166667	1	28.7
IAJD_253	735.104	8.9221	89.93	9	1	32	1	0.837209302	0	3.1
IAJD_263	729.188	12.6974	51.24	6	0	36	1	0.847826087	0	4.03
IAJD_264	759.214	12.0599	71.47	7	1	38	1	0.85106383	0	7.98
IAJD_265	785.296	14.2578	51.24	6	0	40	1	0.86	0	1.85
IAJD_266	815.322	13.6203	71.47	7	1	42	1	0.862745098	0	2.34
IAJD_267	590.89	7.3787	71.47	7	1	26	1	0.8	1	20.1
IAJD_268	646.998	8.9391	71.47	7	1	30	1	0.820512821	0	5.46
IAJD_269	673.08	11.137	51.24	6	0	32	1	0.833333333	0	2.89
IAJD_270	703.106	10.4995	71.47	7	1	34	1	0.837209302	1	17.2
IAJD_271	731.16	11.2797	71.47	7	1	36	1	0.844444444	1	30.8
IAJD_272	787.268	12.8401	71.47	7	1	40	1	0.857142857	1	25.8
IAJD_273	867.439	13.7628	69.7	8	0	49	0	0.981132075	1	26.7
IAJD_274	883.438	12.7352	89.93	9	1	50	0	0.981132075	0	NaN
IAJD_275	683.116	9.8452	60.47	7	0	36	0	0.975609756	0	NaN
IAJD_276	713.142	9.2077	80.7	8	1	38	0	0.976190476	1	61
IAJD_277	725.197	11.0155	60.47	7	0	39	0	0.977272727	0	NaN
IAJD_278	755.223	10.378	80.7	8	1	41	0	0.977777778	1	67.1
IAJD_279	851.44	14.5264	60.47	7	0	48	0	0.981132075	0	0.94
IAJD_280	881.466	13.8889	80.7	8	1	50	0	0.981481481	1	11
IAJD_282	893.521	15.6967	60.47	7	0	51	0	0.982142857	0	NaN
IAJD_283	923.547	15.0592	80.7	8	1	53	0	0.98245614	1	15.5
IAJD_284	935.602	16.867	60.47	7	0	54	0	0.983050847	0	NaN
IAJD_285	965.628	16.2295	80.7	8	1	56	0	0.983333333	0	NaN
IAJD_287	699.115	8.6493	69.7	8	0	34	0	0.975609756	0	NaN

IAJD_288	715.114	7.6217	89.93	9	1	35	0	0.975609756	1	302
IAJD_289	598.954	7.5046	60.47	7	0	30	0	0.971428571	0	NaN
IAJD_290	628.98	6.8671	80.7	8	1	32	0	0.972222222	1	60.5
IAJD_291	657.034	7.9113	69.7	8	0	34	0	0.973684211	0	NaN
IAJD_292	673.033	6.8837	89.93	9	1	35	0	0.973684211	0	NaN
IAJD_294	562.836	6.4239	71.47	7	1	22	1	0.787878788	1	59.4
IAJD_297	562.836	6.4239	71.47	7	1	22	1	0.787878788	0	5.43
IAJD_299	586.899	5.6968	80.7	8	1	29	0	0.96969697	0	1.2
IAJD_300	691.051	9.0191	80.7	8	1	29	1	0.829268293	0	5.15
IAJD_301	735.104	9.0357	89.93	9	1	32	1	0.837209302	1	150
IAJD_308	773.241	12.5636	71.47	7	1	39	1	0.854166667	1	39.9
IAJD_309	817.294	12.5802	80.7	8	1	42	1	0.86	1	27.8
IAJD_310	817.294	12.4666	80.7	8	1	42	1	0.86	1	31.6
IAJD_311	562.836	6.3103	71.47	7	1	22	1	0.787878788	0	0

Table S2 Sensitivity analysis of the Good/Low labeling cutoff.

Threshold	Pos	Neg	Val ROC-AUC	Val PR-AUC	Val balanced Acc	Val MCC	Train CV ROC-AUC	Train CV PR-AUC
1e ⁶	186	45	0.7105	0.8693	0.6725	0.3	0.8258 ± 0.0888	0.9354 ± 0.0447
5e ⁶	149	82	0.8686	0.9316	0.7912	0.6	0.7518 ± 0.0576	0.8460 ± 0.0393
1e ⁷	125	106	0.7491	0.7426	0.6409	0.3	0.7135 ± 0.1010	0.7199 ± 0.0884
5e ⁷	48	183	0.8270	0.4687	0.8149	0.5	0.7530 ± 0.0957	0.4721 ± 0.1134
1e ⁸	17	214	0.7462	0.2008	0.5189	0.02	0.7490 ± 0.0743	0.3227 ± 0.1672

Table S3 Optimized classifier hyperparameters during model selection.

Classifier	Hyperparameter	Value
Linear Regression	Inverse regularization strength (C)	0.1, 1, 10
	Penalty (penalty)	l2
Random Forest (RF)	Number of trees (n_estimators)	100, 300
	Maximum depth (max_depth)	0, 5, 10
Extreme Gradient	Number of trees (n_estimators)	100, 300
Boosting (XGB)	Maximum depth (max_depth)	3, 5, 7
	Learning rate (learning_rate)	0.05, 0.10

Table S4 Optimized classifier hyperparameters during ECFP R and nbits selection.

Classifier	Hyperparameter	Value
Extreme Gradient Boosting (XGB) 2_64	Number of trees (n_estimators)	100
	Maximum depth (max_depth)	3
	Learning rate (learning_rate)	0.007
	Subsample (subsample)	0.95
	Column subsample(colsample_bytree)	0.40
	Minimum child weight	3
	Gamma (gamma)	0.20
	L1 regularization (reg_alpha)	0.001
	L2 regularization (reg_lambda)	2.8
Extreme Gradient Boosting (XGB) 3_256	Number of trees (n_estimators)	50
	Maximum depth (max_depth)	3
	Learning rate (learning_rate)	0.01
	Subsample (subsample)	0.9
	Column subsample (colsample_bytree)	0.8
	Minimum child weight	1
	Gamma (gamma)	0
	L1 regularization (reg_alpha)	0
	L2 regularization (reg_lambda)	2
Extreme Gradient Boosting (XGB) 3_2048	Number of trees (n_estimators)	100
	Maximum depth (max_depth)	5
	Learning rate (learning_rate)	0.01
	Subsample (subsample)	0.7
	Column subsample(colsample_bytree)	0.6
	Minimum child weight	1
	Gamma (gamma)	0.1
	L1 regularization (reg_alpha)	0.2
	L2 regularization (reg_lambda)	2.5

Table S5 Generation diagnostics summary.

Metric	Value
SELFIES / RDKit validity	0.662
Uniqueness	1.000
Novelty vs positives	1.000
NN similarity (mean)	0.54
NN similarity (median)	0.49
NN similarity < 0.85	88.2%

Table S6 Sensitivity analysis of scoring function weights

Weight perturbation	Spearman ρ	Top-10 overlap	Interpretation
w ₁ (P _{active}) +30%	0.973	9/10 (90%)	Highly stable
w ₁ (P _{active}) -30%	0.966	8/10 (80%)	Highly stable
w ₂ (SA) +30%	0.970	8/10 (80%)	Highly stable
w ₂ (SA) -30%	0.970	9/10 (90%)	Highly stable
w ₃ (PAINS) +30%	1.000	10/10 (100%)	Identical ranking
w ₃ (PAINS) -30%	1.000	10/10 (100%)	Identical ranking
w ₄ (QED) +30%	0.979	10/10 (100%)	Highly stable
w ₄ (QED) -30%	0.978	10/10 (100%)	Highly stable
w ₅ (Lipinski) +30%	1.000	10/10 (100%)	Identical ranking
w ₅ (Lipinski) -30%	1.000	10/10 (100%)	Identical ranking

XGB_3_256_code

```
import numpy as np
import pandas as pd
```

```
from scipy.stats import loguniform, uniform, randint
from sklearn.model_selection import StratifiedKFold, RandomizedSearchCV,
cross_validate
from sklearn.metrics import make_scorer, balanced_accuracy_score,
matthews_corrcoef
from xgboost import XGBClassifier
```

```
all_sheets = pd.read_excel("/content/ecfp_features.xlsx", sheet_name=None)
df = all_sheets["r3_n256"]
```

```
y = df["label"].values
X = df.drop(columns=["label", "SMILES"], errors="ignore").values
```

```
print("X shape:", X.shape, "y shape:", y.shape)
vals, cnts = np.unique(y, return_counts=True)
print("Class counts:", dict(zip(vals, cnts)))
```

```
clf = XGBClassifier(
    objective="binary:logistic",
    eval_metric="logloss",
    tree_method="hist",
    random_state=42,
    n_jobs=-1,
    scale_pos_weight=1.0
)
```

```
param_dist = {
    "n_estimators": randint(100, 600),
    "max_depth": randint(2, 7),
    "learning_rate": loguniform(1e-2, 2e-1),
    "subsample": uniform(0.7, 0.3),
    "colsample_bytree": uniform(0.6, 0.4),
    "min_child_weight": loguniform(5e-1, 10),
    "gamma": loguniform(1e-6, 2e-1),
    "reg_alpha": loguniform(1e-8, 5e-1),
    "reg_lambda": loguniform(5e-1, 20),
}
```

```
inner_cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
outer_cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=2024)
```

```
search = RandomizedSearchCV(
    estimator=clf,
    param_distributions=param_dist,
    n_iter=400,
    scoring="roc_auc",
    cv=inner_cv,
    n_jobs=-1,
    random_state=42,
    verbose=0
)
```

```
scoring = {
    "roc_auc": "roc_auc",
    "pr_auc": "average_precision",
}
```

```

        "balanced_acc": make_scorer(balanced_accuracy_score),
        "mcc": make_scorer(matthews_corrcoef),
    }

cv_out = cross_validate(
    search, X, y,
    cv=outer_cv,
    scoring=scoring,
    n_jobs=-1,
    return_train_score=False
)

def mean_sd(a):
    return float(np.mean(a)), float(np.std(a))

roc_m, roc_s = mean_sd(cv_out["test_roc_auc"])
pr_m, pr_s = mean_sd(cv_out["test_pr_auc"])
ba_m, ba_s = mean_sd(cv_out["test_balanced_acc"])
mcc_m, mcc_s = mean_sd(cv_out["test_mcc"])

print("\n==== Nested CV (outer loop) metrics: mean ± SD =====")
print(f"outer-CV ROC-AUC : {roc_m:.4f} ± {roc_s:.4f}")
print(f"outer-CV PR-AUC : {pr_m:.4f} ± {pr_s:.4f}")
print(f"outer-CV BA : {ba_m:.4f} ± {ba_s:.4f}")
print(f"outer-CV MCC : {mcc_m:.4f} ± {mcc_s:.4f}")

```